Experiment 8

```
# ============================================================
# EXPERIMENT 8
# INTERACTIVE SALES AND PROFIT DASHBOARD
# Python + Google Colab
#
# Dataset: SampleSuperstore.csv
# Rows: 9994
# Columns: 13
# ============================================================


# ============================================================
# 1. INSTALL REQUIRED LIBRARIES
# ============================================================

!pip install dash plotly -q


# ============================================================
# 2. IMPORT LIBRARIES
# ============================================================

import pandas as pd
import numpy as np
import plotly.express as px

from dash import Dash, dcc, html, Input, Output
from google.colab import files


# ============================================================
# 3. UPLOAD DATASET
# ============================================================

print("Select SampleSuperstore.csv")

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

print("\nSelected file:", file_name)


# ============================================================
# 4. LOAD DATASET
# ============================================================
```

```
df = pd.read_csv(file_name)

print("\nDataset loaded successfully!")


# ============================================================
# 5. EXPLORE DATASET
# ============================================================

print("\n" + "=" * 60)
print("DATASET INFORMATION")
print("=" * 60)

print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])


print("\nColumns:")
print(df.columns.tolist())


print("\nFirst 5 records:")
display(df.head())


print("\nLast 5 records:")
display(df.tail())


print("\nData types:")
print(df.dtypes)


print("\nMissing values:")
print(df.isnull().sum())


print("\nDuplicate records:")
print(df.duplicated().sum())


print("\nSummary statistics:")
display(df.describe())


# ============================================================
# 6. CLEAN COLUMN NAMES
# ============================================================

df.columns = df.columns.str.strip()
```

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# ============================================================
# 7. CHECK REQUIRED COLUMNS
# ============================================================

required_columns = [
    "Region",
    "Category",
    "Sub-Category",
    "Sales",
    "Profit"
]

for column in required_columns:

    if column not in df.columns:

        raise ValueError(
            f"Required column '{column}' is missing."
        )


print("\nAll required columns are available.")


# ============================================================
# 8. CREATE FILTER VALUES
# ============================================================

regions = sorted(
    df["Region"].dropna().unique()
)

categories = sorted(
    df["Category"].dropna().unique()
)

subcategories = sorted(
    df["Sub-Category"].dropna().unique()
)


# ============================================================
# 9. CREATE DASH APPLICATION
# ============================================================

app = Dash(__name__)
```

```
# ============================================================
# 10. DASHBOARD LAYOUT
# ============================================================

app.layout = html.Div(

    style={
        "fontFamily": "Arial",
        "backgroundColor": "#f5f6fa",
        "padding": "20px"
    },

    children=[

        # ------------------------------------------------------
        # TITLE
        # ------------------------------------------------------

        html.H1(
            "Sales and Profit Dashboard",
            style={
                "textAlign": "center",
                "marginBottom": "30px"
            }
        ),


        # ------------------------------------------------------
        # DESCRIPTION
        # ------------------------------------------------------

        html.P(
            "Interactive analysis of Sales and Profit "
            "using the Sample Superstore dataset.",
            style={
                "textAlign": "center",
                "fontSize": "16px"
            }
        ),


        # ------------------------------------------------------
        # FILTERS
        # ------------------------------------------------------

        html.Div(

            style={
                "display": "flex",
```

```
            "gap": "20px",
            "marginBottom": "25px"
        },

        children=[

            # REGION FILTER
            html.Div(

                style={"flex": "1"},

                children=[

                    html.Label(
                        "Region"
                    ),

                    dcc.Dropdown(

                        id="region-filter",

                        options=[
                            {
                                "label": region,
                                "value": region
                            }

                            for region in regions
                        ],

                        value=regions,

                        multi=True
                    )
                ]
            ),


            # CATEGORY FILTER
            html.Div(

                style={"flex": "1"},

                children=[

                    html.Label(
                        "Category"
                    ),
```

```
                dcc.Dropdown(

                    id="category-filter",

                    options=[
                        {
                            "label": category,
                            "value": category
                        }

                        for category in categories
                    ],

                    value=categories,

                    multi=True
                )
            ]
        ),


        # SUB-CATEGORY FILTER
        html.Div(

            style={"flex": "1"},

            children=[

                html.Label(
                    "Sub-Category"
                ),

                dcc.Dropdown(

                    id="subcategory-filter",

                    options=[
                        {
                            "label": subcategory,
                            "value": subcategory
                        }

                        for subcategory in subcategories
                    ],

                    value=subcategories,

                    multi=True
                )
```

```
                ]
            )
        ]
    ),


    # =========================================================
    # BAR CHART
    # =========================================================

    html.Div(

        style={
            "backgroundColor": "white",
            "padding": "15px",
            "marginBottom": "20px"
        },

        children=[

            dcc.Graph(
                id="sales-category"
            )
        ]
    ),


    # =========================================================
    # SCATTER PLOT
    # =========================================================

    html.Div(

        style={
            "backgroundColor": "white",
            "padding": "15px",
            "marginBottom": "20px"
        },

        children=[

            dcc.Graph(
                id="profit-sales"
            )
        ]
    ),


    # =========================================================
```

```
        # REGION SALES CHART
        # ==================================================

        html.Div(

            style={
                "backgroundColor": "white",
                "padding": "15px"
            },

            children=[

                dcc.Graph(
                    id="region-sales"
                )
            ]
        )
    ]
)


# ============================================================
# 11. DASHBOARD CALLBACK
# ============================================================

@app.callback(

    [

        Output(
            "sales-category",
            "figure"
        ),

        Output(
            "profit-sales",
            "figure"
        ),

        Output(
            "region-sales",
            "figure"
        )
    ],

    [

        Input(
            "region-filter",
```

```
            "value"
        ),

        Input(
            "category-filter",
            "value"
        ),

        Input(
            "subcategory-filter",
            "value"
        )
    ]
)


def update_dashboard(
    selected_regions,
    selected_categories,
    selected_subcategories
):


    # ============================================================
    # FILTER DATA
    # ============================================================

    filtered_df = df[
        df["Region"].isin(
            selected_regions
        )
        &
        df["Category"].isin(
            selected_categories
        )
        &
        df["Sub-Category"].isin(
            selected_subcategories
        )
    ]


    # ============================================================
    # 12. BAR CHART
    # SALES BY CATEGORY
    # ============================================================

    category_sales = (
```

```
        filtered_df

        .groupby(
            [
                "Category",
                "Region"
            ],
            as_index=False
        )

        ["Sales"]

        .sum()
    )


    bar_fig = px.bar(

        category_sales,

        x="Category",

        y="Sales",

        color="Region",

        barmode="group",

        text_auto=".2s",

        title="Sales by Category"
    )


    bar_fig.update_layout(

        xaxis_title="Category",

        yaxis_title="Total Sales",

        legend_title="Region"
    )


    # ============================================================
    # 13. SCATTER PLOT
    # PROFIT VS SALES BY SUB-CATEGORY
    # ============================================================
```

```
scatter_data = (

    filtered_df

    .groupby(
        [
            "Sub-Category",
            "Category"
        ],

        as_index=False
    )

    .agg({

        "Sales": "sum",

        "Profit": "sum"

    })
)


scatter_fig = px.scatter(

    scatter_data,

    x="Sales",

    y="Profit",

    color="Category",

    hover_name="Sub-Category",

    size="Sales",

    title="Profit vs Sales by Sub-Category"
)


scatter_fig.update_layout(

    xaxis_title="Total Sales",

    yaxis_title="Total Profit",

    legend_title="Category"
)
```

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    # ============================================================
    # 14. SALES BY REGION
    # ============================================================

    region_sales = (

        filtered_df

        .groupby(
            "Region",
            as_index=False
        )

        ["Sales"]

        .sum()
    )


    region_fig = px.bar(

        region_sales,

        x="Region",

        y="Sales",

        text_auto=".2s",

        title="Sales by Region"
    )


    region_fig.update_layout(

        xaxis_title="Region",

        yaxis_title="Total Sales"
    )


    # ============================================================
    # 15. RETURN CHARTS
    # ============================================================

    return (
```

```
        bar_fig,

        scatter_fig,

        region_fig

    )


# ============================================================
# 16. RUN DASHBOARD
# ============================================================

print("\nStarting dashboard...")

app.run(
    mode="external",
    debug=False
)


OUTPUT:
   ____________________________________________________________________________
8.9/8.9 MB 93.3 MB/s eta 0:00:00
Select SampleSuperstore.csv
• SampleSuperstore.csv.zip(application/x-zip-compressed) - 167457 bytes, last modified: 10/6/2026 - 100% done
Saving SampleSuperstore.csv.zip to SampleSuperstore.csv.zip

Selected file: SampleSuperstore.csv.zip

Dataset loaded successfully!

============================================================
DATASET INFORMATION
============================================================
Number of rows: 9994
Number of columns: 13

Columns:
['Ship Mode', 'Segment', 'Country', 'City', 'State', 'Postal Code',
'Region', 'Category', 'Sub-Category', 'Sales', 'Quantity', 'Discount',
'Profit']

First 5 records:
```

Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.9600	2	0.00	41.9136

Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.9400	3	0.00	219.5820
Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	14.6200	2	0.00	6.8714
Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.5775	5	0.45	-383.0310
Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies	Storage	22.3680	2	0.20	2.5164

Last 5 records:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
9989	Second Class	Consumer	United States	Miami	Florida	33180	South	Furniture	Furnishings	25.248	3	0.2	4.1028
9990	Standard Class	Consumer	United States	Costa Mesa	California	92627	West	Furniture	Furnishings	91.960	2	0.0	15.6332
9991	Standard Class	Consumer	United States	Costa Mesa	California	92627	West	Technology	Phones	258.576	2	0.2	19.3932
9992	Standard Class	Consumer	United States	Costa Mesa	California	92627	West	Office Supplies	Paper	29.600	4	0.0	13.3200

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
9993	Second Class	Consumer	United States	Westminster	California	92683	West	Office Supplies	Appliances	243.160	2	0.0	72.9480

```
Data types:
Ship Mode        object
Segment          object
Country          object
City             object
State            object
Postal Code       int64
Region           object
Category         object
Sub-Category     object
Sales           float64
Quantity          int64
Discount        float64
Profit          float64
dtype: object

Missing values:
Ship Mode       0
Segment         0
Country         0
City            0
State           0
Postal Code     0
Region          0
Category        0
Sub-Category    0
Sales           0
Quantity        0
Discount        0
Profit          0
dtype: int64

Duplicate records:
17

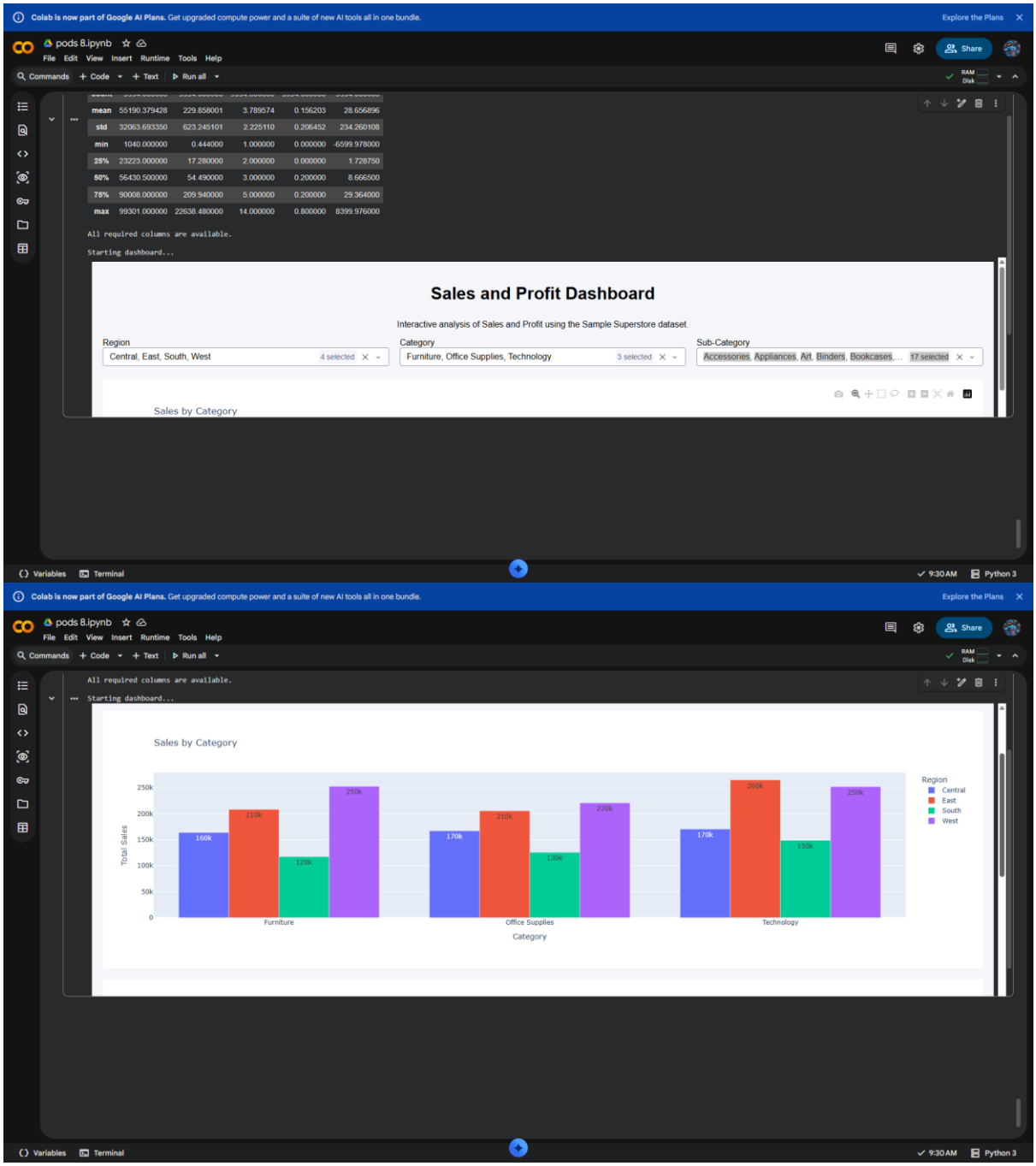
Summary statistics:
```

	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	55190.379428	229.858001	3.789574	0.156203	28.656896
std	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1040.000000	0.444000	1.000000	0.000000	-6599.978000

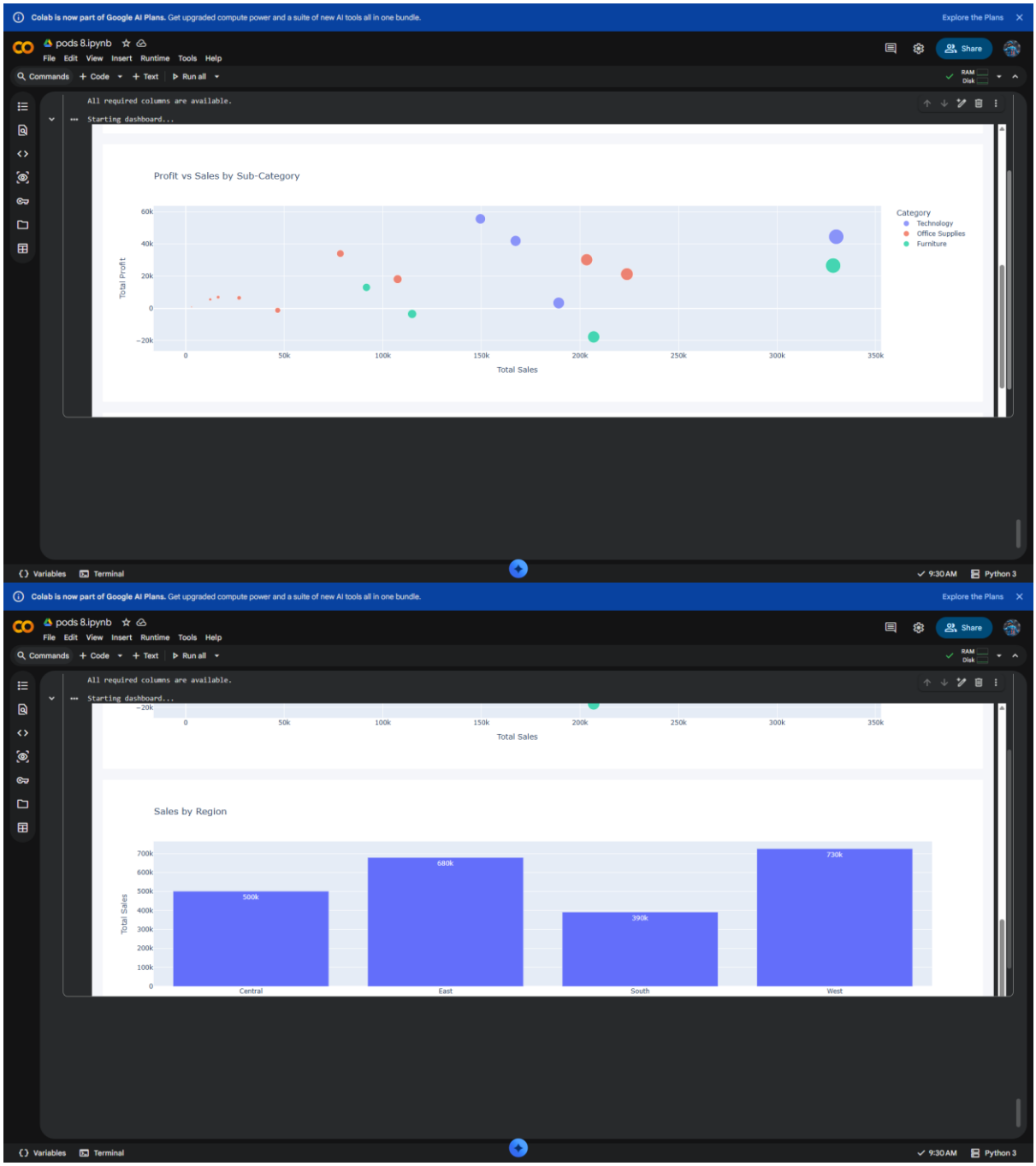
	Postal Code	Sales	Quantity	Discount	Profit
25%	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	90008.000000	209.940000	5.000000	0.200000	29.364000
max	99301.000000	22638.480000	14.000000	0.800000	8399.976000

```
All required columns are available.

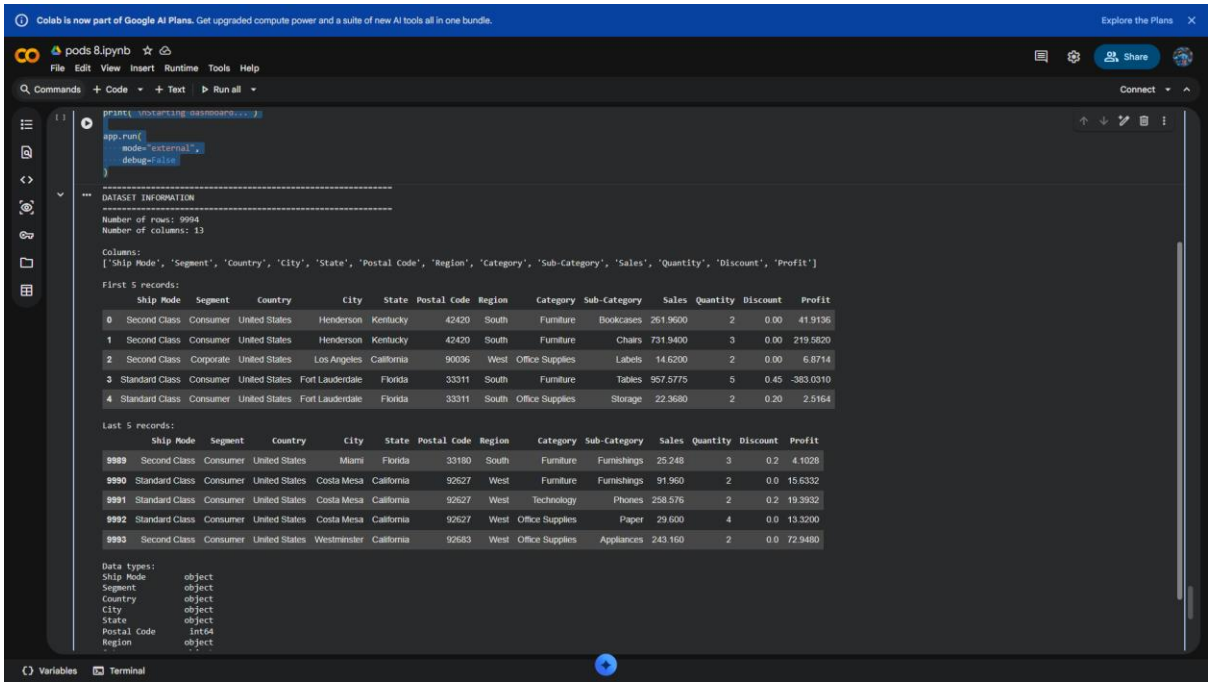
Starting dashboard...
```


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mean 55190.379428 229.858001 3.789574 0.156203 28.656896
std 32063.693350 623.245101 2.225110 0.206452 234.260108
min 1040.000000 0.444000 1.000000 0.000000 -6599.978000
25% 23223.000000 17.280000 2.000000 0.000000 1.728750
50% 56430.500000 54.490000 3.000000 0.200000 8.666500
75% 90008.000000 209.940000 5.000000 0.200000 29.364000
max 99301.000000 22638.480000 14.000000 0.800000 8399.976000
All required columns are available.
Starting dashboard...
Sales and Profit Dashboard
Interactive analysis of Sales and Profit using the Sample Superstore dataset.
Region
Central, East, South, West
4 selected
Category
Furniture, Office Supplies, Technology
3 selected
Sub-Category
Accessories, Appliances, Art, Binders, Bookcases,...
17 selected
Sales by Category
Variables Terminal
9:30 AM Python 3

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All required columns are available.
Starting dashboard...
Sales by Category
Total Sales
250k
200k
150k
100k
50k
0
160k
210k
120k
250k
170k
210k
130k
220k
170k
260k
150k
250k
Furniture
Office Supplies
Technology
Category
Region
Central
East
South
West
Variables Terminal
9:30 AM Python 3


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Share
Commands + Code + Text Run all
All required columns are available.
Starting dashboard...
Profit vs Sales by Sub-Category
Total Profit
60k
40k
20k
0
−20k
0
50k
100k
150k
200k
250k
300k
350k
Total Sales
Category
Technology
Office Supplies
Furniture
Variables
Terminal
9:30 AM
Python 3

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File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
All required columns are available.
Starting dashboard...
−20k
0
50k
100k
150k
200k
250k
300k
350k
Total Sales
Sales by Region
Total Sales
700k
600k
500k
400k
300k
200k
100k
0
500k
680k
390k
730k
Central
East
South
West
Variables
Terminal
9:30 AM
Python 3


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File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
print(\nstarting dashboard...)
app.run(
mode="external",
debug=False
)
DATASET INFORMATION
Number of rows: 9994
Number of columns: 13
Columns:
['Ship Mode', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Category', 'Sub-Category', 'Sales', 'Quantity', 'Discount', 'Profit']
First 5 records:
Ship Mode Segment Country City State Postal Code Region Category Sub-Category Sales Quantity Discount Profit
0 Second Class Consumer United States Henderson Kentucky 42420 South Furniture Bookcases 261.9600 2 0.00 41.9136
1 Second Class Consumer United States Henderson Kentucky 42420 South Furniture Chairs 731.9400 3 0.00 219.5820
2 Second Class Corporate United States Los Angeles California 90036 West Office Supplies Labels 14.6200 2 0.00 6.8714
3 Standard Class Consumer United States Fort Lauderdale Florida 33311 South Furniture Tables 957.5775 5 0.45 -383.0310
4 Standard Class Consumer United States Fort Lauderdale Florida 33311 South Office Supplies Storage 22.3680 2 0.20 2.5164
Last 5 records:
Ship Mode Segment Country City State Postal Code Region Category Sub-Category Sales Quantity Discount Profit
9989 Second Class Consumer United States Miami Florida 33180 South Furniture Furnishings 25.248 3 0.2 4.1028
9990 Standard Class Consumer United States Costa Mesa California 92627 West Furniture Furnishings 91.960 2 0.0 15.6332
9991 Standard Class Consumer United States Costa Mesa California 92627 West Technology Phones 258.576 2 0.2 19.3932
9992 Standard Class Consumer United States Costa Mesa California 92627 West Office Supplies Paper 29.600 4 0.0 13.3200
9993 Second Class Consumer United States Westminster California 92683 West Office Supplies Appliances 243.160 2 0.0 72.9480
Data types:
Ship Mode object
Segment object
Country object
City object
State object
Postal Code int64
Region object
Variables
Terminal